

Announcements

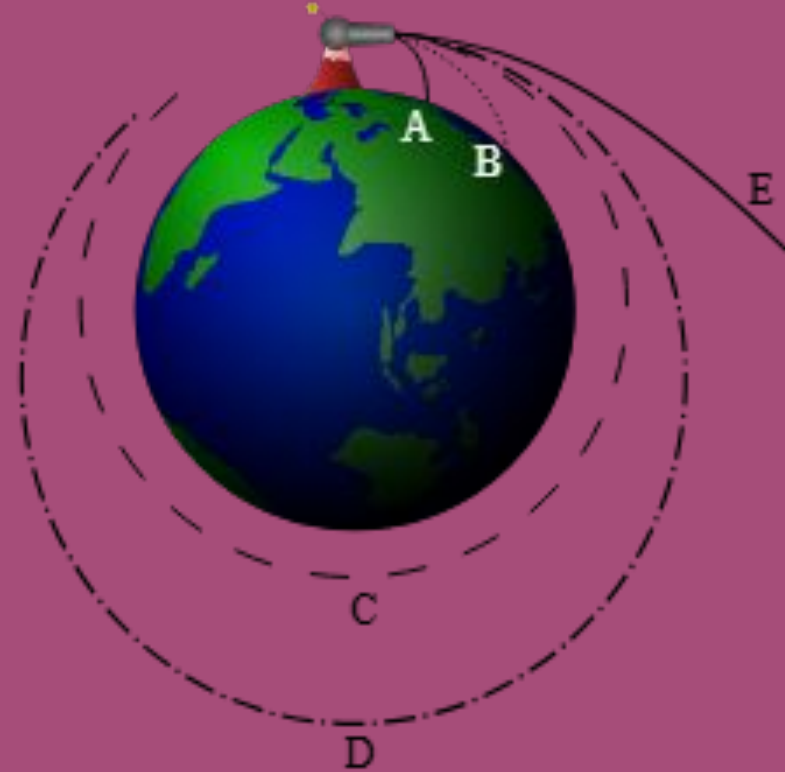
- Next Week is a night lab! Meet in **PSF-647** at **7:30 p.m.**

Orbital Motion

And Kepler's Laws

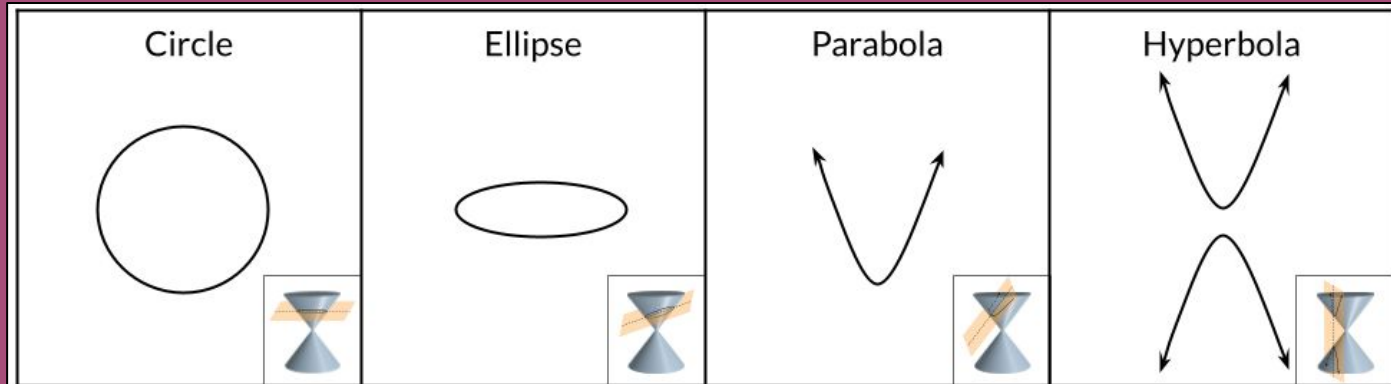
Falling and Missing the Ground

- Gravity is pulling you straight down
- Think about throwing something
 - It travels a distance horizontally as it falls to the ground
 - The faster it goes, the further it goes before hitting the ground
- An orbit is just this, but going so fast that it keeps moving horizontally instead of hitting the ground
 - If there is nothing to noticeably slow it down (like in space), then it can continue forever



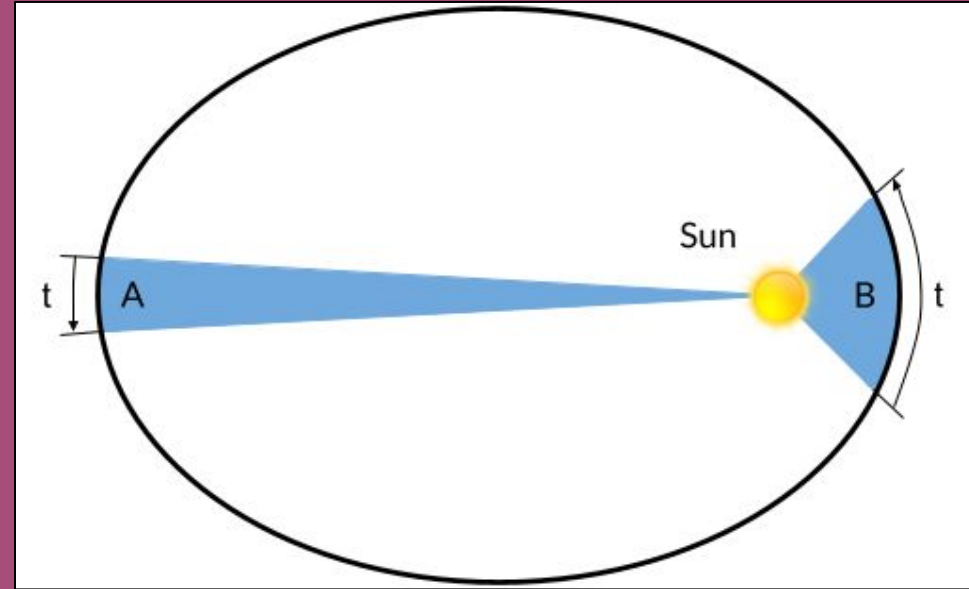
Kepler's 1st Law

- The path taken by an orbiting object is an ellipse
- An ellipse has two foci, the thing being orbited (i.e. the Sun) is at one of them
 - *Technically*, the center of mass of the two objects is at one of the foci, but the Sun is so big we say it is at the middle for our purposes today
- The measure of how “stretched” the ellipse is compared to a perfect circle **eccentricity**
 - Eccentricity of 0 means a perfect circle, 1 is a parabola, greater than 1 is a hyperbola



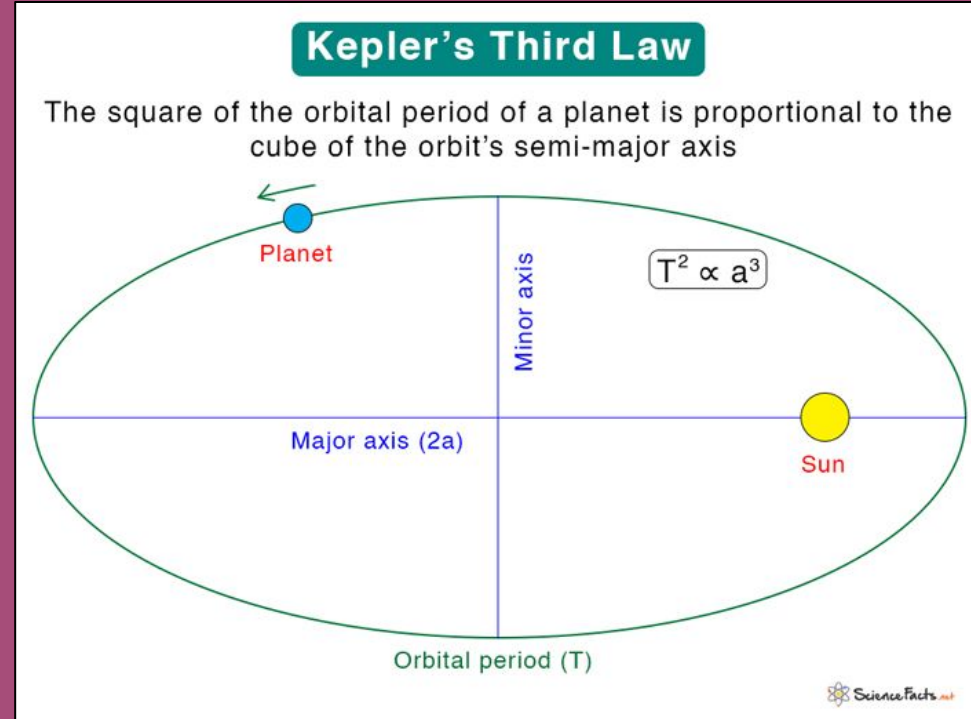
Kepler's 2nd Law

- An imaginary line between the orbiting object and what it is orbiting sweeps out the same area in the same amount of time, regardless of eccentricity
- This means that objects are traveling faster when they are closer to the object being orbited
 - And therefore are slower when far away



Kepler's 3rd Law

- $P^2 \propto a^3$ - Orbital period (P) squared is proportional to the length of the semi-major axis cubed
 - Proportional to means that there is some other constant that needs to be included for them to be equal
 - In the context of objects orbiting the Sun, and when using units of Earth years and astronomical units (AU), it becomes $P^2 = a^3$
 - Sometimes "T" is used for period instead of "P"



Escape Velocity

- Escape velocity is the velocity needed to escape an object's gravity (at a specific distance)
- The formula is:
$$v_e = \sqrt{\frac{2GM}{r}}$$
- If another object's velocity is greater than this, then it is not orbiting but passing by
 - The “orbit” is parabolic or hyperbolic
 - It is not “bound”
- If we see a comet with a velocity greater than the Sun's escape velocity - it is probably an interstellar object like 1I/Oumuamua or 3I/Atlas

↑ Not an alien
spacecraft

Today's Lab

- Not really much to say with this one, just work through the lab
- You will need to use the orbital simulator and DreamPlot that are both linked in the lab
 - You will also need to make a **copy** of the Google Sheets spreadsheet
- Note that the Tables and Graphs Submission today is all digital using the above DreamPlot and spreadsheet
- For questions 13-18 you will need to do some calculations to determine if the comet is bound to the Sun

Image Credits (In order of appearance)

- Wikipedia Newton's Cannonball
 - https://en.wikipedia.org/wiki/Newton%27s_cannonball
- The lab itself
- The lab itself
- Science Facts
 - <https://www.sciencefacts.net/keplers-third-law.html>